

ALUMNI WEB APPLICATION

Network Portal

Software Requirements Specification (SRS)

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Revision History

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Alumni Connect Network Portal. The document is intended for developers, project managers, QA engineers, and stakeholders involved in the design, development, testing, and maintenance of the system. This is Version 1.0, representing the initial baseline release of the platform.

1.2 Project Overview

Alumni Connect is a full-stack web application designed to help educational institutions manage, track, and connect with their alumni community. The system provides a structured platform for institutions to maintain alumni records, approve self-registrations, and allow alumni to network and explore each other's professional journeys.

1.3 Scope

The Alumni Connect portal covers the following functional areas:

- Admin Panel for managing alumni records, approvals, and platform statistics
- Alumni self-registration with institution approval workflow
- Alumni login and self-managed profile editing
- Public-facing alumni directory with search and filter capabilities
- Password management (forgot/reset password via email)
- Responsive web interface for desktop and mobile usage

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
JWT	JSON Web Token — used for stateless authentication
API	Application Programming Interface
CRUD	Create, Read, Update, Delete — basic data operations
UI	User Interface
UX	User Experience
OTP	One-Time Password
REST	Representational State Transfer — architectural style for APIs
Multer	Node.js middleware for handling multipart/form-data (file uploads)

1.5 References

- React Documentation — <https://react.dev>
- Vite Documentation — <https://vitejs.dev>
- Node.js Documentation — <https://nodejs.org>
- Express.js Documentation — <https://expressjs.com>
- MySQL Documentation — <https://dev.mysql.com/doc>
- JSON Web Token RFC 7519

2. Overall Description

2.1 Product Perspective

Alumni Connect is a standalone web application developed using a modern MERN-style stack (React + Node.js + MySQL). It operates as a browser-based portal accessible on any device with an internet connection. The system is self-contained and does not depend on third-party alumni management software, although it may integrate with email services for password reset functionality.

2.2 System Architecture

The system follows a client-server architecture with a clear separation of concerns:

Layer	Technology	Role
Frontend (Client)	React + Vite + Tailwind CSS	Renders UI, handles routing and state
Backend (Server)	Node.js + Express.js	Processes business logic, REST API endpoints
Database	MySQL	Persistent storage for alumni and user data
Authentication	JWT (JSON Web Token)	Stateless session management
File Upload	Multer	Handles profile photo uploads
Email	Nodemailer / SMTP	Password reset email delivery

2.3 User Classes and Characteristics

User Role	Access Level	Primary Activities
Administrator	Full system control	Manage alumni records, approve registrations, view statistics, add/edit/delete alumni
Alumni (Registered)	Profile & directory access	Login, edit own profile, browse alumni directory
Public Visitor	Read-only public pages	View home page, register, request password reset

2.4 Operating Environment

- Browser: Modern browsers (Chrome, Firefox, Edge, Safari — latest two versions)
- Device: Desktop, tablet, and mobile (responsive design)
- Minimum screen width: 360px
- Backend: Node.js LTS version on any major operating system
- Database: MySQL 8.0+
- Hosting: cPanel or equivalent shared/VPS hosting

2.5 Assumptions and Dependencies

- SMTP credentials must be configured for password reset emails to function
- The institution assigns admin credentials; no public admin registration is available
- Profile photos are stored on the server filesystem
- MySQL database must be pre-configured with the required schema before deployment

3. Functional Requirements

3.1 Authentication Module

3.1.1 Admin Login

- The system shall allow admin users to login with a username and password
- A JWT token shall be issued on successful login
- Invalid credentials shall return an appropriate error message
- Admin sessions shall expire after a defined token lifetime

3.1.2 Alumni Login

- Alumni shall log in using their registered email and password
- Only approved alumni accounts shall be permitted to login
- A JWT token shall be issued to authenticated alumni

3.1.3 Forgot / Reset Password

- Alumni shall be able to request a password reset via their registered email
- The system shall send a time-limited reset token to the provided email address
- Alumni shall use the reset link to set a new password
- Used or expired reset tokens shall be invalidated after use

3.2 Alumni Registration

- Public users shall be able to self-register as alumni by providing personal and professional details
- Registration submissions shall be placed in a pending state awaiting admin approval
- Alumni shall receive a confirmation once approved
- Admin shall be able to approve or reject pending registrations

3.3 Admin Dashboard

- The dashboard shall display total alumni count
- The dashboard shall show alumni counts by employment status: Employed, Self-Employed, Studying, Pending
- Quick action links to Add Alumni, Manage Alumni, and Pending Approvals shall be available
- A Recent Alumni table shall show the latest records with name, designation, batch, status, and approval state

3.4 Alumni Management (Admin)

3.4.1 Add Alumni

- Admin shall be able to add alumni profiles directly without requiring self-registration
- Required fields include name, email, designation, batch year, employment status, and company
- Profile photo upload shall be supported

3.4.2 Manage Alumni

- Admin shall view a complete list of all alumni with search and filter capabilities
- Admin shall edit any alumni profile
- Admin shall delete alumni records with confirmation prompt

3.4.3 Pending Approvals

- Admin shall view all self-registered alumni awaiting approval
- Admin shall approve or reject each pending registration

3.5 Alumni Profile Management

- Authenticated alumni shall be able to view and edit their own profile
- Editable fields include profile photo, designation, company, employment status, bio, and contact details
- Changes shall be saved immediately to the database

3.6 Alumni Directory (Connect Page)

- All authenticated alumni shall be able to browse the alumni directory
- Alumni cards shall display profile photo, name, designation, company, batch year, and employment status
- The logged-in user's own card shall be highlighted with an 'Edit My Profile' button
- Filters shall allow alumni to search by name, batch year, employment status, and designation
- The directory shall show total alumni count

3.7 User Management (Admin)

- Admin shall be able to view and manage other admin user accounts
- Admin shall check username availability before creating new users
- Admin shall perform full CRUD operations on admin users

4. Non-Functional Requirements

4.1 Performance

- Page load time for main views shall not exceed 3 seconds on a standard broadband connection
- API response time for read operations shall be under 500ms under normal load
- The system shall support at least 100 concurrent users without degradation

4.2 Security

- All passwords shall be hashed using bcrypt before storage
- JWT tokens shall be signed with a secret key and validated on every protected API request
- Password reset tokens shall expire within 1 hour of generation
- File uploads shall be restricted to image types (JPEG, PNG, GIF, WebP)
- CORS policy shall be configured to allow only approved frontend origins

4.3 Usability

- The UI shall be responsive and functional on screen widths from 360px to 1920px
- Forms shall provide clear validation messages for required or invalid fields
- Navigation shall be consistent and intuitive across all user roles

4.4 Reliability

- The system shall have an uptime target of 99.5% per month
- Database connections shall include error handling and retry logic
- The backend shall log errors with sufficient detail for debugging

4.5 Maintainability

- Code shall follow a modular structure with clear separation between routes, middleware, and frontend pages
- Environment variables shall be used for all configuration (database credentials, SMTP, JWT secret)
- All sensitive information shall be excluded from version control via .env files

5. System Modules

5.1 Backend Module Structure

File / Folder	Purpose
routes/admin.js	Admin login, user management, alumni CRUD via admin
routes/alumni.js	Alumni login, registration, profile edit, forgot/reset password
routes/stats.js	Dashboard statistics (total alumni, career counts)
config/db.js	MySQL database connection configuration
middleware/auth.js	JWT token verification middleware
middleware/upload.js	Multer configuration for profile photo uploads
server.js	Express app entry point, route registration, CORS setup

5.2 Frontend Module Structure

File / Path	Purpose
pages/admin/AdminDashboard.jsx	Admin dashboard with statistics and quick actions
pages/admin/ManageAlumni.jsx	View, edit, and delete alumni records
pages/admin/AddAlumni.jsx	Form to add new alumni records
pages/admin/PendingApprovals.jsx	Review and approve pending self-registrations
pages/admin/ManageUsers.jsx	Manage administrator user accounts
pages/alumni/AlumniEditProfile.jsx	Alumni self-edit profile form
pages/public/Home.jsx	Landing page for public visitors
pages/public/Login.jsx	Admin login page
pages/public/UserLogin.jsx	Alumni login page
pages/public/UserRegistration.jsx	Alumni self-registration form
pages/public/ForgotPassword.jsx	Forgot password request form
pages/public/ResetPassword.jsx	Reset password form (via email token)
pages/user/ViewAlumni.jsx	Alumni directory / connect page
components/Navbar.jsx	Navigation bar component
components/AdminLayout.jsx	Admin panel layout wrapper

5.3 API Endpoint Summary

Method	Endpoint	Description	Auth
POST	/user/login	Admin login — returns JWT	None
GET	/users	List all admin users	Admin JWT
GET	/users/check-username/:username	Check username availability	Admin JWT
POST	/alumni/login	Alumni login — returns JWT	None
POST	/alumni/register	Alumni self-registration	None
GET	/alumni	Get all alumni	JWT
GET	/alumni/:id	Get specific alumni profile	JWT
PUT	/alumni/:id	Update alumni profile	JWT
DELETE	/alumni/:id	Delete alumni record	Admin JWT
PUT	/alumni/:id/approve	Approve or reject alumni	Admin JWT
POST	/alumni/forgot-password	Send password reset email	None
POST	/alumni/reset-password	Reset password via token	None
GET	/total-alumni	Total alumni count	Admin JWT
GET	/careers	Distinct designations count	Admin JWT

6. Data Requirements

6.1 Alumni Data Entity

Field	Type	Description
id	INT (PK)	Auto-increment primary key
name	VARCHAR(100)	Full name of the alumni
email	VARCHAR(150)	Unique email address
password	VARCHAR(255)	Bcrypt-hashed password
designation	VARCHAR(100)	Current job title/designation
company	VARCHAR(100)	Current employer
batch_year	YEAR	Year of graduation
employment_status	ENUM	Employed / Self-Employed / Studying / Unemployed
profile_photo	VARCHAR(255)	File path to uploaded profile photo
is_approved	BOOLEAN	Admin approval status
reset_token	VARCHAR(255)	Password reset token
reset_token_expiry	DATETIME	Expiry time of reset token
created_at	DATETIME	Record creation timestamp

6.2 Admin User Data Entity

Field	Type	Description
id	INT (PK)	Auto-increment primary key
username	VARCHAR(50)	Unique administrator username
password	VARCHAR(255)	Bcrypt-hashed password
role	VARCHAR(20)	User role (default: administrator)
created_at	DATETIME	Account creation timestamp

7. Interface Requirements

7.1 User Interface

- The frontend shall be built using React 18+ with Vite as the build tool
- Tailwind CSS shall be used for consistent and responsive styling
- The admin panel shall use Admin Layout as a consistent wrapper with sidebar navigation
- The alumni portal shall use a top navigation bar
- All forms shall include client-side validation before submission
- The alumni directory cards shall display profile photo, name, designation, company, batch, and status

7.2 Software Interfaces

- Frontend communicates with the backend via RESTful HTTP/HTTPS API calls
- Axios or Fetch API shall be used on the frontend for HTTP requests
- The backend shall connect to MySQL via mysql2 or sequelize ORM
- Email delivery shall use Nodemailer with SMTP configuration

7.3 Communication Interfaces

- All API communications shall use HTTPS in production
- JWT tokens shall be transmitted in HTTP Authorization headers (Bearer scheme)
- File uploads shall use multipart/form-data encoding

8. Constraints and Limitations (v1.0)

- No real-time features (live chat, notifications) are included in v1.0
- Alumni cannot message each other directly through the portal in v1.0
- No multi-institution support; the system serves a single institution
- Profile photos are stored on the server filesystem, not cloud storage
- No mobile native app is provided; the web app is mobile-responsive
- No automated email verification during alumni registration; approval is manual by admin

9. Future Plans (Roadmap)

The following features are planned for future versions of Alumni Connect. These are not part of the v1.0 release but represent the product roadmap based on institutional needs and user feedback.

9.1 Version 2.0 — Communication & Engagement

9.1.1 Alumni Messaging System

- Private messaging between alumni within the portal
- Message notifications via email or in-app alerts
- Conversation history and inbox management

9.1.2 Event Management Module

- Admin can create and publish alumni events (reunions, webinars, meetups)
- Alumni can register/RSVP for events through the portal
- Event calendar view with reminders

9.1.3 Announcements & News Feed

- Admin can post institutional announcements and news
- Alumni see a personalized feed on their dashboard

9.2 Version 2.1 — Professional Networking

9.2.1 Job Board

- Alumni and admin can post job openings
- Alumni can apply and track application status
- Email notifications for new job posts matching alumni profile

9.2.2 Mentorship Program

- Senior alumni can register as mentors
- Junior alumni can request mentorship and schedule sessions
- Mentorship tracking and feedback collection

9.2.3 Enhanced Alumni Profiles

- LinkedIn-style profile with skills, achievements, publications, and portfolio links
- Educational background section (multiple degrees)
- Endorsements and recommendations between alumni

9.3 Version 2.2 — Analytics & Reporting

- Admin dashboard with advanced analytics: employment trends, batch-wise statistics, career growth charts

- Exportable reports (PDF / Excel) for institutional accreditation and records
- Alumni engagement metrics (logins, profile updates, event participation)
- Visualized data with charts and graphs using Recharts or Chart.js

9.4 Version 3.0 — Platform Expansion

9.4.1 Multi-Institution Support

- Support for multiple educational institutions on a single platform (SaaS model)
- Institution-specific branding, domain, and admin panel
- Super-admin role to manage all institutions

9.4.2 Mobile Application

- React Native mobile app for Android and iOS
- Push notifications for messages, events, and announcements
- Camera integration for profile photo capture

9.4.3 Cloud Storage Integration

- Profile photos and documents stored on AWS S3 or Cloudinary
- Scalable file storage replacing local filesystem

9.4.4 Social Login

- Login with Google, LinkedIn, or Facebook OAuth
- Auto-populate alumni profile from LinkedIn data

9.5 Technical Infrastructure Improvements

- Migrate to cloud hosting (AWS, DigitalOcean, or Azure) for scalability
- Implement Redis caching for improved API response performance
- Add CI/CD pipeline with GitHub Actions for automated testing and deployment
- Email verification during registration (OTP or link-based)
- Two-factor authentication (2FA) for admin accounts
- API rate limiting and DDoS protection

10. Acceptance Criteria

Feature	Acceptance Criteria
Admin Login	Admin can login with valid credentials and is redirected to dashboard. Invalid credentials return error.
Alumni Registration	Registered alumni appear in Pending Approvals. Admin can approve or reject.
Alumni Login	Approved alumni can login. Pending/rejected alumni see appropriate message.
Forgot Password	Reset email is received within 2 minutes. Token expires after 1 hour.
Admin Dashboard	Correct counts displayed for Total, Employed, Self-Employed, Studying, and Pending.
Add Alumni	Admin-added alumni appear immediately in the directory after saving.
Edit Alumni	Changes saved by admin or alumni are reflected immediately on reload.
Delete Alumni	Deleted alumni are removed from the directory and cannot login.
Alumni Directory	All approved alumni are displayed. Filters function correctly.
Profile Photo Upload	Photo uploads succeed for JPEG/PNG files. Non-image files are rejected.
Responsive Design	All pages render correctly at 360px, 768px, 1024px, and 1920px widths.

11. Appendix

11.1 Technology Stack Summary

Category	Technology	Version
Frontend Framework	React	18+
Frontend Build Tool	Vite	5+
CSS Framework	Tailwind CSS	3+
Backend Runtime	Node.js	LTS (20+)
Backend Framework	Express.js	4+
Database	MySQL	8.0+
ORM / DB Driver	mysql2	Latest
Authentication	jsonwebtoken	Latest
Password Hashing	bcrypt	Latest
File Upload	Multer	Latest
Email	Nodemailer	Latest
HTTP Client (FE)	Axios / Fetch API	—

11.2 Document Sign-off

Role	Name	Date	Signature
Project Owner	Sunsys Technologoies		
Lead Developer	Sudhanandhini S		
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Institution Representative			

End of Document